

LUWUK BUBUNG, Indonesia

WMO: 970860

Lat: 1.0414S

Lon: 122.7711E

Elev: 2

StdP: 101.30

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	23.8	17.1	12.2	25.5	18.2	13.1	25.8	8.3	26.6	7.4	27.1	2.3	280	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	32.5	26.4	32.1	26.4	31.7	26.3	27.2	31.1	27.1	31.0	26.8	30.7	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	30.2	25.8	21.0	30.0	25.6	20.8	29.9	86.1	31.4	85.1	31.2	84.2	31.0	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.1	6.1	5.3	DB	22.1	33.7	0.9	0.8	21.5	34.2	20.9	34.7	20.5	35.1	19.8	35.7
			WB	18.4	28.0	1.2	0.6	17.5	28.4	16.8	28.8	16.1	29.1	15.2	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	28.5	28.4	28.3	27.9	27.6	26.9	26.5	26.5	27.1	28.0	28.4	28.5	(6)
	DBStd	1.12	0.85	0.86	0.83	0.77	0.89	0.92	0.88	0.90	0.82	0.71	0.71	0.81	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6473	574	516	568	537	547	508	511	513	514	559	553	574	(10)
	CDD18.3	3431	316	282	309	287	289	258	252	254	264	301	303	316	(11)
	CDH23.3	37548	3813	3333	3633	3239	3129	2490	2200	2243	2647	3423	3604	3792	(12)
(13)	CDH26.7	11812	1461	1224	1309	1049	903	530	359	396	647	1141	1338	1454	(13)
(14)	Wind	WSAvg	2.3	1.8	1.9	1.9	2.3	2.6	2.9	3.2	3.1	2.6	2.0	1.7	(14)
(15) Precipitation	PrecAvg	1457	94	101	145	154	148	220	196	114	60	44	74	107	(15)
	PrecMax	2439	230	309	308	324	288	821	494	374	164	127	193	284	(16)
	PrecMin	728	3	8	28	67	61	80	80	0	5	0	16	49	(17)
	PrecStd	317	54	62	75	55	53	146	105	87	47	43	41	49	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.0	32.7	32.1	31.7	31.0	30.2	30.3	31.0	31.8	32.6	32.8	(19)
		MCWB	26.1	26.2	26.7	26.7	26.8	26.1	25.4	25.0	25.6	25.9	26.5	26.5	(20)
	2%	DB	32.2	32.1	32.0	31.5	31.1	30.2	29.7	29.7	30.4	31.3	32.0	32.2	(21)
		MCWB	26.3	26.3	26.5	26.5	26.4	25.7	25.1	24.8	25.2	25.8	26.3	26.5	(22)
	5%	DB	31.7	31.6	31.5	30.9	30.5	29.8	29.1	29.2	30.0	31.0	31.5	31.7	(23)
		MCWB	26.3	26.2	26.3	26.4	26.2	25.5	24.9	24.5	25.0	25.7	26.2	26.5	(24)
	10%	DB	31.2	31.1	31.0	30.4	30.0	29.2	28.6	28.7	29.5	30.5	31.0	31.2	(25)
		MCWB	26.2	26.1	26.2	26.1	26.0	25.3	24.6	24.3	24.8	25.6	26.1	26.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.2	27.5	27.3	27.2	26.7	26.2	26.0	26.4	26.8	27.2	27.6	(27)
		MCDB	31.0	31.1	31.5	30.9	30.8	29.9	29.4	29.3	29.9	30.6	31.4	31.5	(28)
	2%	WB	27.0	27.0	27.1	27.0	26.8	26.2	25.7	25.5	26.0	26.5	26.8	27.2	(29)
		MCDB	31.1	31.1	31.0	30.6	30.3	29.5	28.9	28.9	29.6	30.5	30.9	31.2	(30)
	5%	WB	26.7	26.6	26.7	26.6	26.5	25.9	25.3	25.1	25.6	26.2	26.6	26.9	(31)
		MCDB	30.8	30.8	30.6	30.2	30.0	29.2	28.4	28.5	29.2	30.2	30.8	30.9	(32)
	10%	WB	26.5	26.3	26.4	26.3	26.2	25.5	24.9	24.6	25.2	25.9	26.3	26.6	(33)
		MCDB	30.5	30.4	30.3	29.8	29.5	28.6	27.9	28.0	28.8	30.0	30.4	30.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.3	5.3	5.1	4.7	4.2	4.1	4.2	4.7	5.2	5.4	5.5	(35)
		MCDBR	6.0	5.9	5.9	5.6	5.2	4.9	4.7	4.8	5.1	5.5	5.9	6.1	(36)
	5% WB	MCWBR	3.3	3.3	3.1	3.0	3.1	3.1	3.1	3.3	3.6	3.7	3.4	3.2	(37)
		MCWBR	5.5	5.6	5.5	5.3	5.0	4.6	4.5	4.6	4.9	5.3	5.6	5.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.415	0.417	0.420	0.410	0.403	0.413	0.412	0.419	0.422	0.457	0.438	0.421	(40)
		taud	2.474	2.463	2.458	2.493	2.528	2.512	2.513	2.473	2.449	2.328	2.397	2.464	(41)
	Ebn at Noon		917	917	906	893	875	853	860	873	890	870	889	906	(42)
		Edh at Noon	115	118	118	111	103	102	104	111	117	133	124	116	(43)
(44) All-Sky Solar Radiation	RadAvg		5.18	5.33	5.35	5.20	4.76	4.30	4.36	4.84	5.70	5.88	5.53	5.12	(44)
	RadStd		0.21	0.37	0.31	0.26	0.33	0.36	0.47	0.45	0.40	0.34	0.28	0.32	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page